Practice workbook · Describe the products and purposes of mitosis

What mitosis produces, drawing daughter cells or nuclei, cell cycle growth and division phases, uses of mitosis and contrast with meiosis.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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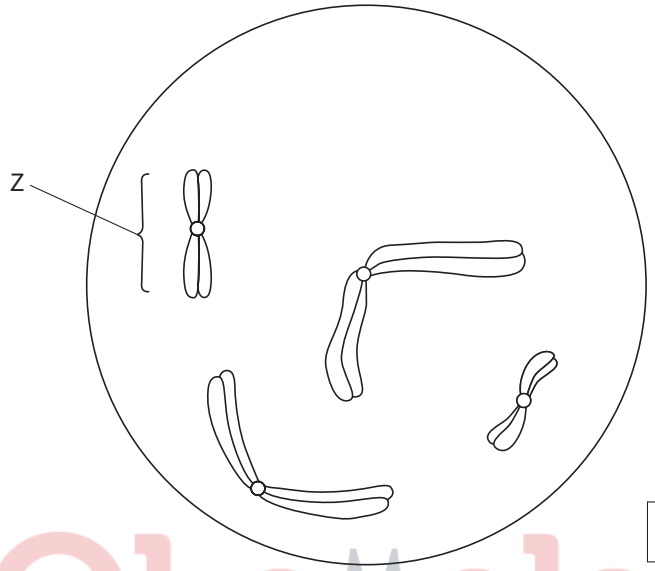
Type ID: HBI-DESCRIBE-MITOSIS-PRODUCTS-AND-USES

1. Question 7 · part c · 2 marks

November 2021 Paper 1

Worked-example retry · Attempt part c only.

7 The diagram shows a cell in early stages of mitosis.



(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

(2)

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

(3)

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1. Question 7 · part c · 2 marks

November 2021 Paper 1

Worked-example retry · Attempt part c only.

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(b) (i) Which one of these molecules would be found in structure Z?

(1)

- ☒ **A** DNA
- ☒ **B** mRNA
- ☒ **C** tRNA
- ☒ **D** ribose sugar

(ii) Calculate the length of structure Z in micrometres.

[1 micrometre = 1×10^{-6} m]

(3)

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length of structure Z = ... micrometres

(c) Describe what is produced when a cell undergoes mitosis.

(2)

.....

.....

.....

.....

(Total for Question 7 = 11 marks)

1. Question 7 · part c · 2 marks · Working and answer

Working and answer

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2. Question 6 · part a.ii · 2 marks

November 2020 Paper 1R

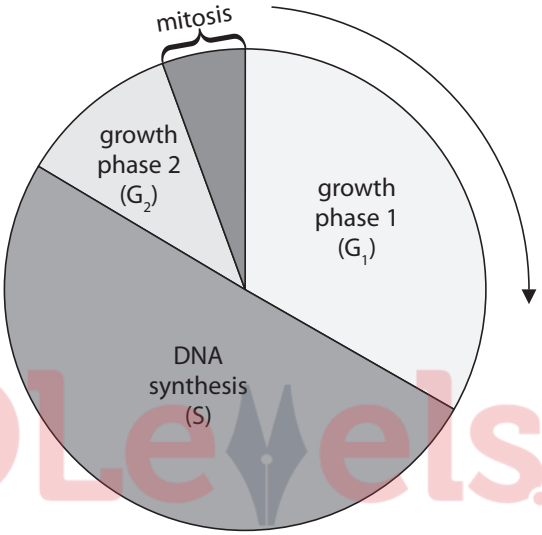
Further practice · Attempt part a.ii only.

6 (a) Cells in the body go through several cell cycles in their lifetime.

Each cell cycle is divided into different stages.

Different events take place in the cell during each stage.

The diagram shows one cell cycle of a body cell.



(i) One cell cycle takes 24 hours to complete.

The total time spent in G_1 and G_2 is ten hours.

Calculate the time spent in mitosis.

(2)

time = ... hours

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P 6 2 0 5 2 A 0 1 2 2 4

2. Question 6 · part a.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

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(ii) Each of the phases of the cell cycle prepare the cell for mitosis.

Suggest the events that take place in the growth phases of the cell cycle.

(2)

(iii) The cell cycle completes by producing two new body cells.

Describe the genetic features of the two cells produced by mitosis.

(2)

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(iv) The table lists some of the stages of mitosis.

Complete the table by giving details of the event that takes place in each stage.

(3)

Stage	Events
metaphase	
anaphase	
telophase	

2. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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3. Question 6 · part a.iii · 2 marks

November 2020 Paper 1R

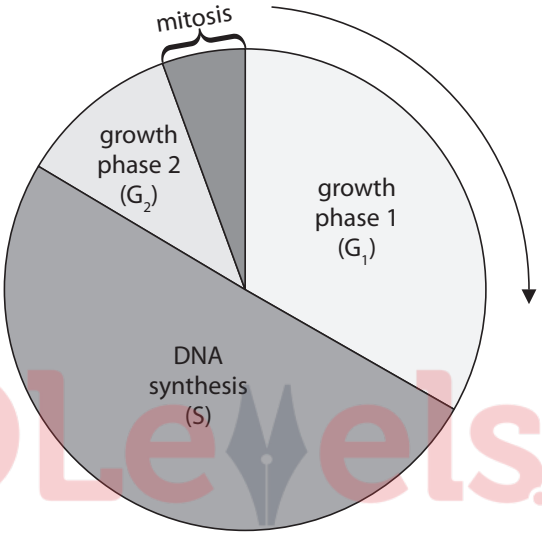
Further practice · Attempt part a.iii only.

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(2)

time = ... hours

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3. Question 6 · part a.iii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iii only.

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(2)

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(2)

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(3)

Stage	Events
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anaphase	
telophase	

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3. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

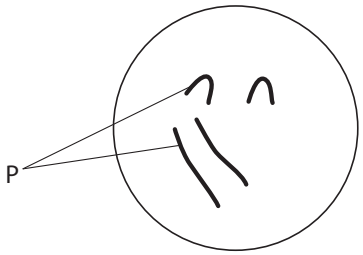
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4. Question 8 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

8 (a) The diagram shows the nucleus of a cell.



(i) Name the structures labelled P.

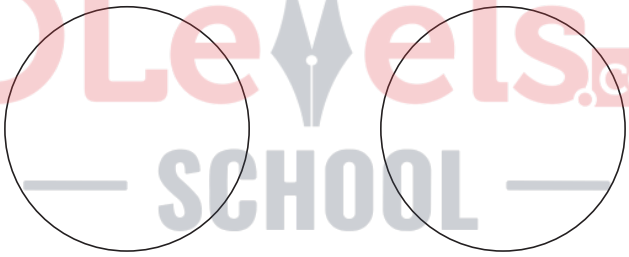
(1)

.....

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

(2)



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

(1)

Stage	Number
complementary bases pair up	
DNA double helix unwinds	
strands separate	
two DNA helices form	

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4. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

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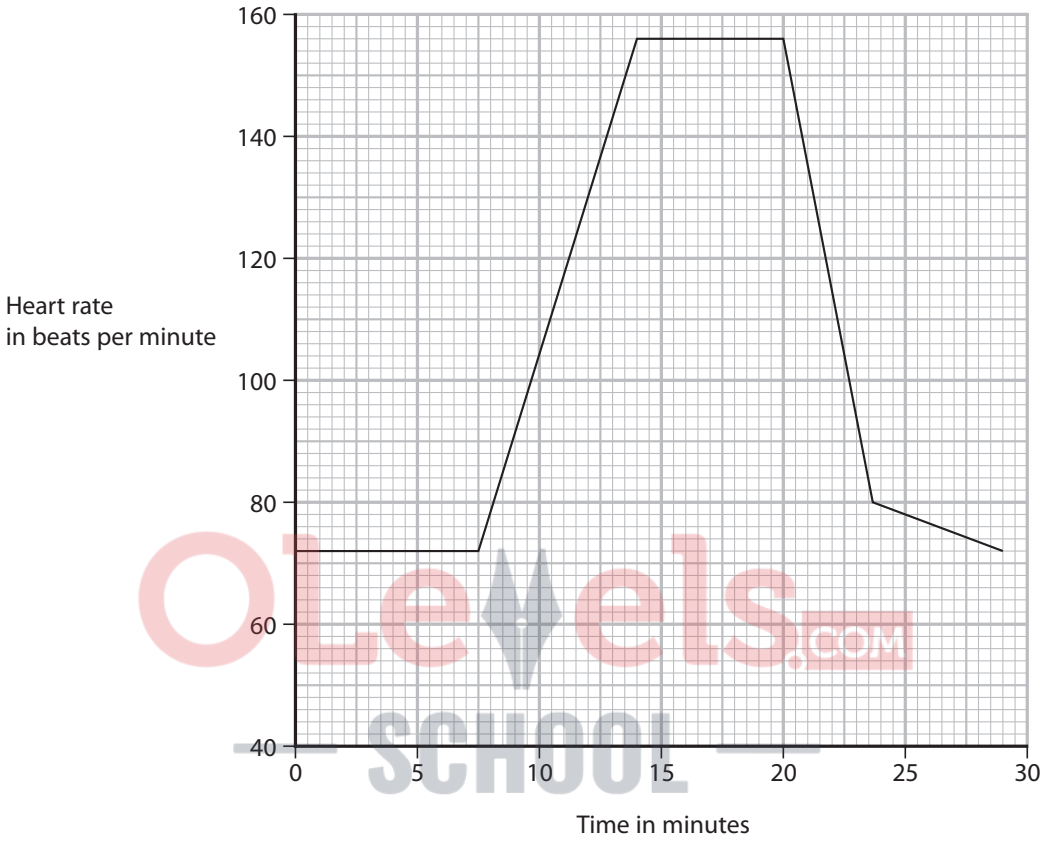
5. Question 2 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

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2 (a) The graph shows how a person's heart rate changes with exercise.



(i) Determine the difference between the maximum heart rate and the resting heart rate.

(2)

difference = ... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

(1)

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

5. Question 2 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

(iii) Complete the passage about what happens in the body during exercise using the correct words.

(4)

During exercise ..... gas moves from the lungs into the blood by a process called ..... . This gas is needed by cells for ..... respiration which releases energy. Also during exercise, the amount of urine produced ..... as water is lost in sweat.

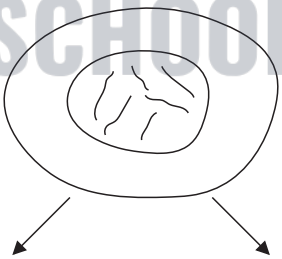
(b) (i) During exercise, body cells get damaged.

Body cells are replaced by a process called mitosis.

The diagram represents one body cell but shows only 6 of the 46 chromosomes.

Complete the diagram to show the new body cells produced from this cell by mitosis.

(2)



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P 7 0 9 7 3 A 0 6 2 4

5. Question 2 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

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(ii) Mitosis takes place in four stages.

Which of these gives the stage of mitosis where chromosomes align at the equator of the cell?

(1)

- ☒ **A** anaphase
- ☒ **B** metaphase
- ☒ **C** prophase
- ☒ **D** telophase

(Total for Question 2 = 10 marks)

Working and answer

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6. Question 4 · part b.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.ii only.

4 The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

(1)

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

(1)

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

(3)

Differences	DNA	Named nucleic acid
1		
2		
3		

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6. Question 4 · part b.ii · 2 marks

November 2023 Paper 1

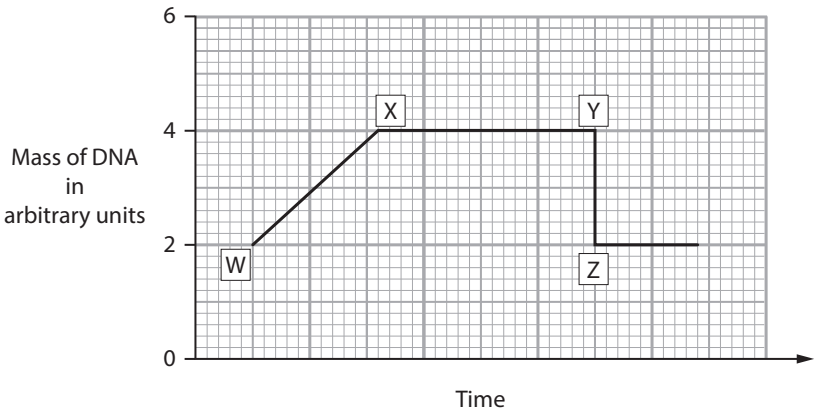
Further practice · Attempt part b.ii only.

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(b) The graph shows the changes in the mass of DNA in a body cell, during mitosis.



(i) Describe the process that produces the change in the mass of DNA between points W and X.

(4)

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(ii) Describe what is happening in the cell between points Y and Z.

(2)

6. Question 4 · part b.ii · 2 marks

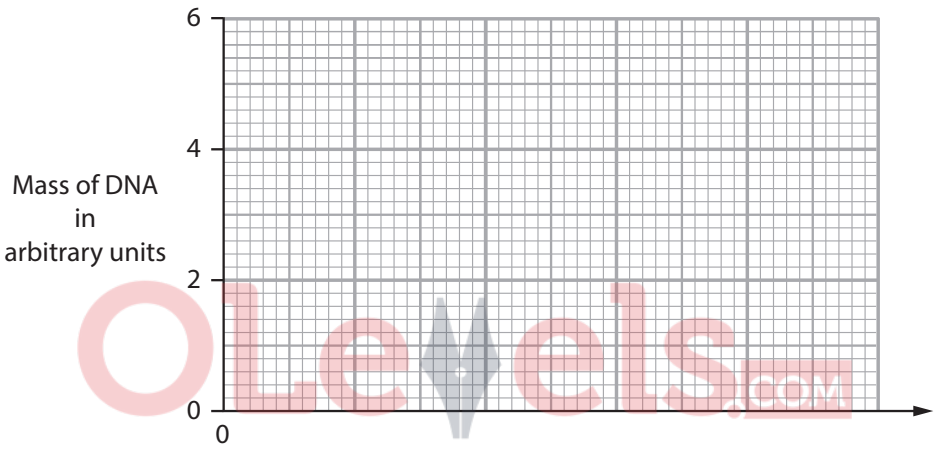
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Further practice · Attempt part b.ii only.

(iii) The production of gametes involves the process of meiosis.

On the grid draw a line to show the changes in the mass of DNA during meiosis.

(3)



(Total for Question 4 = 14 marks)

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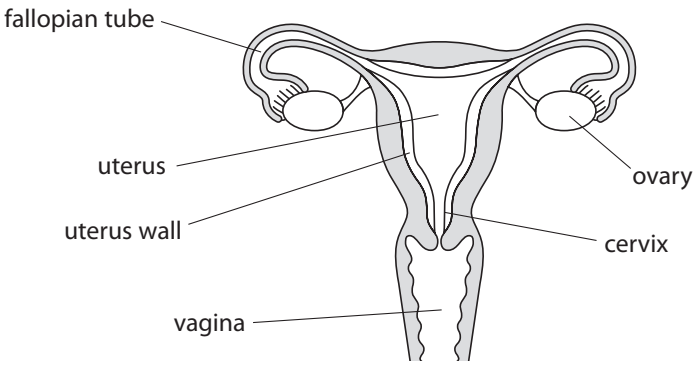
Working and answer

7. Question 2 · part b.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

2 (a) The diagram shows the female reproductive system.



(i) Complete the table by placing one tick (✓) in each row to show the role of each part of the female reproductive system.

(3)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
fallopian tube			
ovary			
vagina			

(ii) Describe the action of the muscles in the uterus wall during labour.

(2)

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7. Question 2 · part b.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

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(iii) Fertilisation takes place in the female reproductive system.

Which of these describes fertilisation?

(1)

- ☒ **A** a sperm cell fuses with an egg cell to form a haploid cell
- ☒ **B** a sperm cell fuses with an egg cell to form an embryo
- ☒ **C** a sperm cell fuses with an egg cell to form a zygote
- ☒ **D** a sperm cell fuses with an egg cell to form a baby

(iv) Before fertilisation can take place, ovulation needs to occur.

Name the hormone that causes ovulation.

(1)

.....

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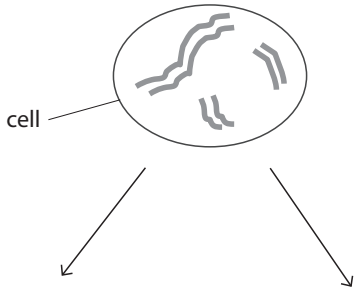
7. Question 2 · part b.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

(b) Following fertilisation and during pregnancy, cell division by mitosis takes place.

The diagram shows a cell with only three pairs of chromosomes instead of a full set.



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(i) Complete the diagram to show the cells formed when this cell divides by mitosis.

(2)

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7. Question 2 · part b.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

(ii) In humans, cell division occurs by mitosis or meiosis.

Draw one straight line from each type of cell division to the cells produced.

(2)

Type of cell division	Cells produced
	two cells, each with 23 chromosomes
mitosis	four cells, each with 46 chromosomes
meiosis	two cells, each with 46 chromosomes
	four cells, each with 23 chromosomes

(Total for Question 2 = 11 marks)

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7. Question 2 · part b.i · 2 marks · Working and answer

Working and answer

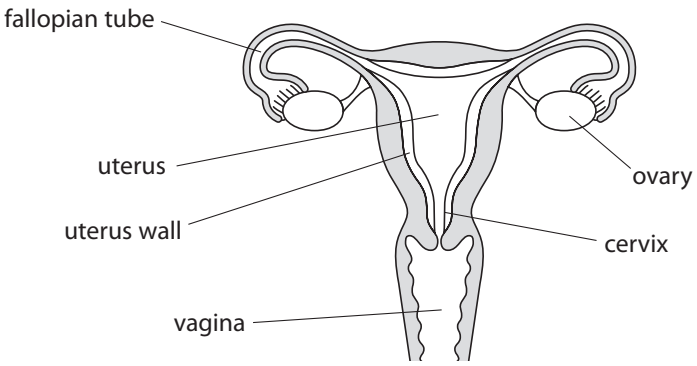
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8. Question 2 · part b.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

2 (a) The diagram shows the female reproductive system.



(i) Complete the table by placing one tick (✓) in each row to show the role of each part of the female reproductive system.

(3)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
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ovary			
vagina			

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(2)

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8. Question 2 · part b.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

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(iv) Before fertilisation can take place, ovulation needs to occur.

Name the hormone that causes ovulation.

(1)

.....

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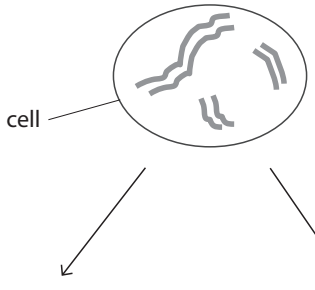
8. Question 2 · part b.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

(b) Following fertilisation and during pregnancy, cell division by mitosis takes place.

The diagram shows a cell with only three pairs of chromosomes instead of a full set.



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(i) Complete the diagram to show the cells formed when this cell divides by mitosis.

(2)

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8. Question 2 · part b.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) In humans, cell division occurs by mitosis or meiosis.

Draw one straight line from each type of cell division to the cells produced.

(2)

Type of cell division	Cells produced
	two cells, each with 23 chromosomes
mitosis	four cells, each with 46 chromosomes
meiosis	two cells, each with 46 chromosomes
	four cells, each with 23 chromosomes

(Total for Question 2 = 11 marks)

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8. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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9. Question 2 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

2 (a) Cells are made up of different components. Each component has a particular function.

Identify the cell component that transports substances inside the cell.

(1)

- ☒ **A** cell membrane
- ☒ **B** cytoplasm
- ☒ **C** endoplasmic reticulum
- ☒ **D** ribosomes

(b) Mitosis is a type of cell division.

A student uses a light microscope to view a cell undergoing four stages in mitosis.

The diagrams, W, X, Y and Z, show the student's observations.



W X Y Z

(Source: https://www.pinterest.co.uk/pin/812547957739062607/)

(i) Give the letters to show the correct order of the diagrams in the process of mitosis.

(1)

.....

(ii) State one function of mitosis in body cells.

(1)

.....

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9. Question 2 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

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(c) Meiosis is another type of cell division.

Give two differences between the cells produced by meiosis and the cells produced by mitosis.

(2)

1 .....

.....

2 .....

.....

(Total for Question 2 = 5 marks)

Working and answer

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